

Full-scale, advanced-level project specification for a **complete Bubble.io application**, written as **user requirements** with detailed features, workflows, logic, data structures, permissions, and integration expectations.

This is suitable as a **capstone project** for an advanced Bubble.io course.

Full-Scale Advanced Project: Multi-Tenant Project Management & Collaboration SaaS

Project Name:

CollabSphere — Multi-Tenant Project & Team Collaboration Platform

Project Summary

Build a full SaaS platform where **multiple organizations** can manage projects, tasks, users, documents, chats, analytics, and custom workflows.

The platform must support:

- Multi-tenant isolation
- Real-time collaboration
- API integrations
- Role-based permissions
- Dynamic dashboards
- Automation rules
- Fully responsive design

1. User Types & Permissions

1.1 User Roles

Each user belongs to an Organization and has a role:

Owner

- Full access to everything
- Billing & subscription management

- Invite & remove users

Admin

- Manage projects, tasks, teams
- Create custom workflows & automations
- View analytics

Member

- Access assigned projects
- Create/update own tasks
- Collaborate via chat & comments

Client (Optional External Role)

- Read-only access to shared project sections
- Leave comments on deliverables

2. Onboarding & Organization Setup

2.1 Organization Creation

- User creates a new organization via signup form.
- Organization record is automatically created.
- Default pipeline stages, templates, and settings are generated via backend workflow.

2.2 User Invitation Flow

- Owner/Admin enters email → Bubble sends invitation email with access link.
- Invitation tokens expire after X days.
- New users are assigned a role during onboarding.

3. Project Management Module

3.1 Create & Manage Projects

Each project includes:

- Title
- Description
- Owner
- Team members
- Pipeline stage
- Start & end dates
- Status (Active, Completed, Archived)

3.2 Kanban Board

Features:

- Drag-and-drop project stages
- Reorder tasks using draggable elements
- Real-time updates via auto-refresh

3.3 Task Management

Tasks include:

- Title, description
- Assigned to
- Priority (Low, Medium, High, Critical)
- Deadline
- Subtasks
- Attachments
- Comments
- Activity logs (auto-logged on every update)

4. Collaboration & Communication

4.1 Real-Time Chat

- One-to-one chat between users
- Group chat inside each project
- Active typing indicator
- Last-seen status
- Read receipts

4.2 Comments & Mentions

- Comment on tasks, documents, deliverables
- @mention users → triggers notification
- Threaded replies

5. File & Document Management

5.1 Document Library

- Upload files (PDF, images, docs)
- Organize into folders per project
- Auto-versioning system:
 - Re-uploads create new versions
 - View version history
- Access control based on roles

5.2 File Previews

- Preview images, PDFs, videos directly inside app
- “Share public preview link” (expires after X hours)

6. Automation & Workflows (Advanced Logic)

Users can create **IF/THEN automation rules**:

Examples:

- IF task marked completed → THEN notify project owner
- IF deadline approaching in 24h → THEN send reminder
- IF client comments → THEN alert task assignee

Automation builder must include:

- Trigger
- Condition
- Action
- Optional delay / schedule
- Logs of executed automation
- Toggle switch (On/Off)

7. Analytics & Dashboards

7.1 Project Dashboard

- Total projects
- Tasks due today
- Overdue tasks
- Active users
- Completion rates
- Task breakdown by priority

7.2 Organization-Level Analytics

- Usage stats (active users, storage used, API calls)
- Team performance metrics
- Time-tracked reports

7.3 Chart Types

- Line graphs
- Bar charts
- Pie charts
- Trend indicators

(Use Chart plugins + Bubble's calculated fields.)

8. Multi-Tenant Architecture Requirements

8.1 Data Isolation Rules

- Users see only data belonging to their Organization
- All searches must filter by `Organization` field

8.2 Privacy Rules

- Organization = Current User's Org

- Role controls access to admin features
- Client role has limited read-only access

9. API Integrations

9.1 Email Service

- Integrate Postmark / SendGrid for:
 - Invitations
 - Notifications
 - Activity summaries

9.2 External AI Integration (Optional Enhancement)

- Summarize project updates
- Generate meeting notes
- Suggest task breakdowns

9.3 External Calendar Sync

- Google Calendar or Outlook
- Sync project due dates
- Enable 2-way sync for admins

10. Public REST API Exposure

Platform should provide an API layer for external access:

- `/api/projects`
- `/api/tasks`
- `/api/users`

Features:

- API keys per organization
- Rate limiting
- API logs
- API documentation page generated dynamically

11. Billing & Subscription

11.1 Features

- Subscription plans: Starter, Business, Enterprise
- Plan limits:
 - Number of users
 - Storage
 - Automations
 - Analytics access

11.2 Integrations

- Stripe subscription billing
- Webhooks to update subscription status
- Billing history page
- Usage meter dashboard

12. App-Wide Features

12.1 Search & Global Navigation

- Global search bar
- Search for: projects, tasks, users, files

12.2 Notifications

- Real-time notifications
- Email + in-app
- Notification panel & read/unread status

12.3 Mobile-Responsive Layout

- Collapsible sidebar
 - Stacked views for mobile
 - Touch-friendly Kanban drag-drop
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13. Security Requirements

- Enforce strong password policies
- Privacy rules deeply validated
- Data protection at record and field level
- Log sensitive events (login, role change, deletion)
- Session timeout controls

14. Performance Requirements

- Use option sets where possible
- Use backend workflows for heavy operations
- Use “optimized for speed” searches
- Minimal use of recursive workflows
- Cache pages using custom states

15. Deliverables

Students must submit:

1. **Live working Bubble app URL**
2. **Editor read-only link**
3. **Documentation** including:
 - Database structure
 - Workflows overview
 - API integration details
 - Privacy rule design
 - Automation examples
4. **Demo video (5–10 minutes)**